Aiywah FSM

Complete Product Blueprint

Oct 10, 2026

@Digit-us Operations

The definitive product specification for a versatile, industry-agnostic field service management platform - designed to win today and remain relevant for the next decade.

Vision and philosophy

Aiywah FSM exists for one reason: every company that sends people to do work at a location - fix equipment, inspect sites, service clients, install systems - should have a single platform that tracks the entire journey from "we need help" to "the job is done, here's proof, how did we do?" and everything in between.

The platform must serve companies across facility management, HVAC, MEP contracting, equipment services, property maintenance, IT infrastructure, cleaning, biomedical engineering, solar energy, elevator maintenance, and any industry where field work happens. It must be the same product for a 10-person AC repair company in Sharjah and a 500-technician facility management firm in Mumbai - different in scale and configuration, identical in architecture.

Five principles make the platform timeless:

1. **The work, not the tool, is the centre.** Every screen exists to move a job forward. If a feature does not directly help someone create a request, assign it, do the work, prove it was done, bill for it, or improve next time, it does not belong. The platform is not a dashboard people look at - it is a workflow people move through.
2. **WhatsApp is the interface for the field.** Field technicians do not open web portals. They live on WhatsApp. Every notification, every job alert, every daily summary, every client update flows through the channel people already use. The web app is for managers; WhatsApp is for the field and the client.
3. **Proof is enforced, not requested.** GPS-stamped punch-in, mandatory before/after photos, customer signature - these are not optional uploads. The system will not let a job close until the proof exists. This eliminates disputes, builds client trust, and creates an evidence trail that no spreadsheet can match.
4. **Configuration replaces customisation.** The platform does not change its code for different industries. It changes its configuration - service types, form fields, checklist templates, SLA rules, approval chains, branding. A single codebase serves every client; settings make it theirs.
5. **API-first, mobile-ready, cloud-native.** Every feature exists as an API endpoint first, a web view second, and a mobile screen third. This means the mobile app is never an afterthought, integrations are never bolt-ons, and the platform can connect to any ERP, CRM, or accounting system without architectural changes.

Platform architecture

The platform is built as a web application with a REST API layer that serves both the web frontend and the mobile app. Every feature is an API endpoint first - the web views consume the same API that the mobile app and third-party integrations use.

Web application (Laravel 12, PHP 8.2+): The primary interface for administrators, managers, dispatchers, and office staff. Server-side rendered Blade views with Vite for asset compilation. Responsive design that works on desktop and tablet.

REST API (Laravel Sanctum): Token-based authentication. Every module - service requests, assets, contracts, punches, scheduling, notifications - exposes JSON endpoints. API versioning (v1, v2) ensures backward compatibility. Webhook events fire on major state changes so external systems can react.

Mobile app (Flutter or React Native): Native iOS and Android app for field technicians. Optimised for punch in/out, camera (before/after photos), GPS capture, offline job queue, push notifications, and customer signature. Syncs with the server when connectivity returns. The mobile app consumes the same REST API - no separate backend.

Database (MySQL): Single database with tenant_id scoping for multi-tenancy. All tenant data isolated via global Eloquent scopes. Master data (roles, permissions, service category templates) seeded per tenant on provisioning.

File storage (S3 / Cloudflare R2): Tenant-prefixed paths for complete data isolation. Photos, signatures, quotes, invoices, and report PDFs stored with signed URLs for secure access.

Queue and cache (Redis): Background jobs for WhatsApp notifications, daily summaries, PM schedule processing, report generation, and bulk imports. Redis-backed session and cache for performance.

WhatsApp (Meta Cloud API): Integrated into the core notification engine. Per-tenant credentials for Enterprise plans, shared platform number for Starter/Professional plans with company name injection.

Multi-tenancy model: Shared database with tenant_id column on every data table. Subdomain routing ({slug}.aiywah.com). BelongsToTenant Eloquent trait auto-scopes every query. Platform admin panel above all tenants for system management, analytics, and support. Subscription billing via Stripe or regional payment gateways.

User roles across industries

Eight generalised roles cover every service company's structure. The names adapt per industry - a "Technician" in HVAC is a "Doctor" in healthcare QMS, a "Field Agent" in cleaning - but the permission structure and workflow position remain identical.

Role	What they do	Industry examples
Platform Admin	Manages the platform itself - all tenants, billing, system health	Aiywah FSM internal team
Company Admin	Full control within their tenant - users, settings, branding, master data	Owner, GM, IT Manager
Operations Manager	Reviews requests, approves work, assigns teams, does QC, sees all reports	Head of Projects, Branch Manager, Ops Director
Team Lead / Supervisor	Manages a team of field staff. Receives approved work, picks the right technician, monitors team workload	Service Engineer, Shift Supervisor, Area Manager
Field Technician	The person who does the work on-site - accepts jobs, sets ETA, punches in/out, uploads proof, logs expenses	Maintenance Lead, Doctor, Cleaning Operative, IT Engineer

Role	What they do	Industry examples
Front Desk / Dispatcher	First point of contact - logs requests, registers clients, creates service requests, dispatches urgent jobs	Receptionist, Call Centre Agent, Helpdesk Operator
Finance / Accounts	Handles out-of-warranty quotes, generates invoices, tracks payments, reconciles expenses	Accountant, Billing Clerk, AR Team
Client User	External - the customer. Views their requests, raises new ones, checks SLA, downloads invoices, rates service	Building Manager, Hospital Admin, Homeowner

Every role's permissions are managed through a configurable RBAC matrix - 25+ permission keys, 7 access levels (full, none, record-level, grantable, view-only, view-own, edit-own), and dynamic permission extensions that let admins grant individual users access beyond their role's default. Nothing is hardcoded - an admin can give a Front Desk user the ability to approve requests or view analytics without changing any code.

The Team Lead role is optional. Smaller companies skip it - the Operations Manager assigns directly to Technicians. Larger companies use it to add a team management layer (e.g., an Electrical team lead managing 8 electricians, a Plumbing lead managing 5 plumbers).

Module 1: Service request lifecycle

The service request (SR) is the core entity. Everything else - assets, contracts, schedules, invoices - exists to serve the SR lifecycle. A request moves through defined statuses, each transition triggering notifications and logging an audit entry.

Statuses (9 primary)

1. Pending - created by front desk or client portal, waiting for manager review
2. Quoted - out-of-warranty/out-of-contract; price quote sent to client
3. Approved - manager approved (in-warranty) or client accepted the quote
4. Assigned - team lead and/or technician assigned, ETA set
5. In progress - technician punched in on-site, GPS and timestamp captured
6. Pending review - work complete, photos and signature uploaded, awaiting QC
7. Rework - QC failed, sent back with mandatory reason, original timestamps preserved
8. Completed - QC passed, invoice raised if applicable, feedback requested
9. Cancelled - rejected with mandatory reason at any stage

The happy path: Pending → Approved → Assigned → In progress → Pending review → Completed

The out-of-warranty branch: Pending → Quoted → (client accepts) → Approved → continues as normal

The QC failure loop: Pending review → Rework → (technician fixes) → Pending review → Completed

Phase 1 - Request creation. Front desk, client portal, or API creates the SR. System generates a unique reference, stamps the creator, logs the activity, and fires a WhatsApp notification to all involved parties. The

SR captures: client, site/project, service category, service domain, priority, description, attachments, and optionally the specific asset being serviced and the contract it falls under.

Phase 2 - Review and approval. Operations manager reviews the request. The system auto-checks: is this site covered by an active contract? If yes, which contract and what SLA applies? If no active contract, route to finance for a quote. Manager can approve (in-warranty/contract), forward for quoting (out-of-warranty), or reject with a reason.

Phase 3 - Quotation (out-of-warranty only). Finance prepares a quote - either auto-generated from the system's line-item builder or uploaded as a PDF from external accounting software. Quote is sent to the client via WhatsApp with a secure link. Client accepts or rejects through the link. On acceptance, the SR returns to the approval queue.

Phase 4 - Dispatch and assignment. Manager selects the team lead (by service type or area) and/or the specific technician. Smart scheduling suggests the best match by skill, location, availability, and current workload. Technician receives a WhatsApp notification with job details. They review, set an ETA, and accept. Client receives a WhatsApp with the technician's name, ETA, and a live tracking link.

Phase 5 - Field execution. Technician punches in on arrival - system captures exact time and GPS coordinates. Before/after photos are mandatory. Customer signs a digital acceptance form. Optional: expenses logged with receipt photos. Punch out is locked until all proof is uploaded. On punch out, total on-site time is calculated. Status moves to Pending review.

Phase 6 - Quality check. Manager reviews all proof - photos, GPS data, timestamps, expense receipts, customer signature. Pass: in-warranty jobs marked Completed; out-of-warranty jobs routed to finance for invoicing. Fail: sent back as Rework with mandatory notes; technician re-does the work.

Phase 7 - Invoicing (out-of-warranty only). Finance generates or uploads an invoice. Invoice sent to client via WhatsApp. Manager approves the invoice. Status moves to Completed.

Phase 8 - Feedback and closure. Client receives a WhatsApp link to rate the service (star rating + NPS 0–10 + optional comment). Score is linked to the technician, the service type, and the site. Dashboards update in real time. The next morning, staff receive their daily WhatsApp summary.

Module 2: Asset and equipment management

Every service request should be linkable to a specific piece of equipment - not just a location. The asset registry tracks individual equipment throughout its lifecycle.

Asset record fields: Name, asset code (unique tag/barcode), model, serial number, manufacturer, installation date, warranty expiry, location within the site (e.g., "3rd floor, server room B"), service category, status (active/inactive/decommissioned), QR code for field scanning, notes, and photo.

Asset-to-SR linking: When creating an SR, after selecting the site, the user sees a dropdown of assets at that site. Selecting an asset auto-fills the service category, shows the asset's warranty status, and displays its recent service history as a tooltip. The SR record carries the asset_id.

Asset 360° view: A single page showing everything about an asset - its identity (model, serial, manufacturer), its current status, its warranty/contract coverage, every SR ever raised against it (full service history), upcoming PM schedules, total cost of service over its lifetime, and photos from past visits.

QR code scanning (mobile app): Each asset gets a generated QR code. The technician scans it on-site to instantly pull up the asset record, create a new SR pre-filled with client/site/asset, or check-in to a PM job. This eliminates data entry errors and speeds up the field workflow.

Bulk import: CSV/Excel import for assets - asset code, name, site code, category, model, serial number, manufacturer, install date, warranty expiry. Validation preview before committing. Downloadable template file.

Why this matters: An HVAC company's client calls about "the chiller in building 7." Without asset tracking, the SR says "chiller issue at building 7." With it, the SR links to asset CHR-007 (York YVAA chiller, installed 2019, warranty expired 2024, serviced 6 times in 2025, last filter change 45 days ago). That context changes how the technician prepares for the job.

Module 3: Preventive maintenance

Reactive work (something broke → fix it) is only half of field service. The other half is preventive maintenance - scheduled, recurring work that keeps equipment healthy and contracts fulfilled.

PM schedule definition: An admin creates a schedule tied to a site and optionally a specific asset. They set: service category, frequency (daily, weekly, biweekly, monthly, quarterly, biannual, annual), default priority, default description/checklist, and optional default technician assignment. Multiple schedules per site are normal - monthly AC filter changes, quarterly elevator inspections, annual fire safety checks, all running independently.

Auto-SR generation: A daily scheduled job (runs at 6:00 AM) checks all active PM schedules. For any schedule where next_due_date is today or has passed, the system automatically creates a new SR with status Pending, linked to the correct site, asset, category, and contract. If a default technician is set, the SR is auto-assigned. The next_due_date is then recalculated based on the frequency.

PM calendar view: A visual monthly calendar showing all upcoming PM jobs (colour-coded by category or site) alongside reactive SRs that have ETAs set. Each day cell shows the count of scheduled items. Click a day to expand and see the details. Click a job to open the SR.

PM compliance tracking: Dashboard widgets showing: PM jobs completed on time vs overdue, PM completion rate by site, by category, and by technician. This matters for contract compliance - many AMCs require a minimum number of preventive visits per quarter, and falling behind means contract breach.

PM vs reactive split: Reports that show the ratio of preventive to reactive work over time. A healthy service operation has more PM than reactive - it means problems are being prevented. If reactive work is increasing, something in the PM program is failing.

How different industries use PM

- FM companies: monthly AC filter changes, quarterly fire extinguisher inspections, annual elevator certifications
- HVAC companies: seasonal pre-cooling and pre-heating service runs across all client sites
- Equipment companies: usage-based maintenance (every 1,000 hours of generator runtime)
- Property management: weekly common area inspections, monthly pest control visits

- Biomedical: annual calibration of hospital equipment per regulatory requirements

Module 4: Contract and AMC management

Contracts define the rules of engagement between the service company and their client - what is covered, what SLA applies, how many service calls are included, when the contract expires, and what it costs.

Contract record fields: Contract number, client, contract type (AMC, warranty, ad-hoc, defect liability period), start date, end date, covered sites (multi-select from client's projects), covered service categories (multi-select), response SLA in hours (overrides the global SLA matrix when set), maximum service calls per period (optional cap), calls used so far (auto-counted), annual contract value, renewal status (active, expiring soon, expired, renewed, cancelled), auto-renew flag, payment terms, and notes.

Automatic contract matching: When an SR is created, the system checks: does this client have an active contract that covers this site AND this service category? If yes, the SR is auto-linked to the contract, inherits the contract's SLA, and is flagged as in-coverage. If no matching contract exists, the SR is treated as out-of-coverage and routed for quoting. If the contract has a call limit and it has been reached, the system flags this to the manager for a decision - bill as out-of-coverage or honour as goodwill.

Renewal management: A scheduled job checks contracts expiring in 30, 60, and 90 days and creates alerts for the admin. A dashboard widget shows the renewal pipeline - contracts sorted by expiry date, with contract value and client name. This is where revenue retention happens.

Contract performance view: For each contract - SLA compliance percentage, number of calls used vs included, total cost of service delivered vs contract value (is this contract profitable?), client satisfaction score from feedback on SRs under this contract.

How this changes the workflow: Without contract management, every SR gets the same default SLA and the warranty check is a simple date comparison. With it, each SR inherits its own SLA, call limits are tracked, and the finance team can see which contracts are profitable and which are being overserviced.

Module 5: Dispatch and scheduling

Getting the right technician to the right job at the right time is the operational core.

Smart assignment factors: When the manager is assigning a job, the system suggests the best technician based on: skill match (the technician handles this service category), proximity (nearest to the job site based on last known location or home base), availability (not already booked at the proposed time), current workload (fewest active jobs), and past performance (first-time fix rate for this type of work). The manager can override the suggestion.

Calendar view: A visual weekly/monthly calendar showing all scheduled work - both PM jobs (auto-generated) and reactive SRs (with ETAs). Filter by technician, team, service category, or site. Click any slot to see the job details. Drag and drop to reschedule - the system auto-creates a reschedule record, updates the ETA, and notifies the client via WhatsApp.

Workload dashboard: A panel showing each technician's upcoming 7-day schedule with a capacity indicator - green (light), amber (moderate), red (overloaded). Helps the manager make balanced assignments instead of overloading the nearest available person every time.

Dispatch urgency handling: For emergency requests (e.g., water leak, power failure), the manager can flag an SR as urgent. Urgent SRs appear at the top of every queue, trigger an immediate WhatsApp to the assigned technician (bypassing the normal notification schedule), and carry a tighter SLA.

Route optimisation (future): When a technician has multiple jobs in a day, suggest the optimal visiting order to minimise driving time. This is a Phase 3 feature that integrates with mapping APIs.

Module 6: Field execution

This is where the physical work happens - the technician on-site, doing the job, capturing proof.

Punch in: Technician arrives at the site and presses "Punch in" on their phone (mobile app) or the web portal. The system records the exact timestamp and GPS coordinates. The client's site location is compared to the punch-in GPS to verify the technician is actually at the right place. Status changes to In Progress. WhatsApp notification fires to the client ("your technician has arrived") and to internal staff.

Photo capture (mandatory): Before starting work, the technician takes "before" photos showing the current state. After completing work, they take "after" photos showing the result. The system will not allow punch-out until both sets are uploaded. Photos are stored with the SR, timestamped, and visible during QC review.

Configurable service form: Each service category can have a custom checklist or form that the technician fills during the job. An AC service might require: compressor pressure reading, refrigerant level, filter condition (clean/dirty/replaced), thermostat calibration, and drain line check. An elevator inspection might require: cable condition, brake test result, door alignment, emergency phone test. These forms are configured by the admin per service category - the technician sees only the form relevant to their current job.

Customer signature: Before punching out, the technician captures the customer's digital signature on the device screen, confirming the work was done to their satisfaction. The signature is stored as an image with GPS coordinates of the signing location.

Expense logging: The technician can log expenses incurred during the job - parts purchased, travel costs, consumables. Each expense has a category (from the admin-configured list), quantity, unit rate, total, and an optional receipt photo. Expenses are visible to the manager during QC and to finance during invoicing.

Punch out (locked until proof is complete): The punch-out button is disabled until: before photos uploaded, after photos uploaded, customer signature captured, and (if configured) the service form completed. This is the enforcement mechanism - there is no way to close a job without proof. On punch-out, total on-site time is calculated automatically. Status moves to Pending Review.

Offline mode (mobile app): In areas with poor connectivity, the technician can punch in, take photos, fill forms, and capture signatures while offline. Data is queued locally on the device and syncs to the server when connectivity returns. The technician sees a sync indicator showing pending uploads.

Hold and resume: During a job, the technician can place it on hold (e.g., waiting for a part, need to return tomorrow). The system pauses the timer, records the hold reason, and notifies the manager. When the

technician returns, they resume - the timer picks up where it left off. Total hold time is tracked separately from work time.

Module 7: Billing and finance

The financial side closes the service loop - from quoting before the work to invoicing after it, with full expense tracking in between.

Quotation builder: For out-of-coverage work, finance builds a quote using the system's line-item editor - description, quantity, unit rate, amount per line, subtotal, tax, discount, and total. The quote is auto-branded with the company's logo, name, and contact details. Alternatively, finance can upload a PDF quote created in external accounting software. Either way, the quote is sent to the client via WhatsApp with a secure link where they can accept or reject it.

Invoice generation: After QC approval on an out-of-coverage SR, finance generates an invoice using the same line-item editor. The invoice can pull expense data from the technician's field entries - parts used, materials, and labour time. Auto-generated invoices carry the company branding, a unique invoice number, payment terms, and bank/payment details. Invoices are sent to the client via WhatsApp. Manager approval is required before the invoice is marked as submitted.

Payment tracking: Each invoice carries a payment status - unpaid, partially paid, paid. Finance can record payments received (amount, date, method, reference). An aging report shows overdue invoices by 30/60/90 days. WhatsApp reminders can be triggered for overdue payments.

Expense reconciliation: All expenses logged by technicians during field work flow into the expense ledger. Finance can review, approve, or reject individual expense items. The ledger is filterable by project, technician, category, and date range. Export to Excel for integration with external accounting systems.

Revenue reporting: Dashboard widgets showing: revenue by client, revenue by contract type, revenue by service category, and the profitability of each contract (service cost vs contract value). This tells the business which clients and which service types are making money and which are losing it.

Integration path: The REST API exposes invoice and payment data so external accounting systems (QuickBooks, Zoho Books, Tally, Xero) can pull completed invoices and push payment confirmations. Webhook events fire on invoice creation, submission, and payment receipt.

Module 8: Communication engine

The platform communicates through four channels, with WhatsApp as the primary.

WhatsApp (Meta Cloud API): The backbone of all client and field communication. Pre-approved message templates fire automatically at every SR status transition - request received, technician assigned (with name and ETA), work started, work completed, invoice ready, feedback request. Templates use parameters for dynamic data (client name, SR reference, technician name, ETA). Each message is logged with delivery status (sent, delivered, read, failed) and can be retried individually or in bulk.

Email: Secondary channel for formal communications - welcome emails for new users, quote and invoice PDFs as attachments, weekly digest reports for managers, and password reset links. Email templates are branded per tenant.

Push notifications (mobile app): Real-time alerts for technicians - new job assigned, schedule change, urgent dispatch, QC result. Push works alongside WhatsApp, not instead of it. The mobile app handles token registration with Firebase Cloud Messaging.

In-app notifications: A notification bell on every web portal screen showing unread items - new SRs assigned, QC results, approval requests, system alerts. Each notification links directly to the relevant SR or action page. Scoped by role - each user sees only notifications relevant to their work.

Notification preferences (per user): Each user can configure which events trigger notifications, on which channel (WhatsApp, email, in-app, push, or none), and at what frequency (instant, hourly digest, daily digest). A high-volume operations manager might want instant WhatsApp only for QC failures and SLA breaches, with everything else as a daily digest. A technician gets instant WhatsApp for new job assignments and nothing else.

Daily WhatsApp summary: Every morning at 8:00 AM, configured staff receive a WhatsApp message with a secure link (expires after 24 hours) showing a role-specific summary of the previous day - jobs completed, sites visited, approvals made, SLA performance, expense totals. Configured centrally by the admin - who receives it, what they see, and on which days.

Notification log and retry: Every outbound message (WhatsApp, email, push) is logged with timestamp, recipient, template used, delivery status, and API response. Failed messages can be retried individually or in bulk. The log is searchable and exportable for audit purposes.

Module 9: Analytics and reporting

Data without reports is a missed opportunity. Every stakeholder needs different numbers in different formats.

Role-specific dashboards: Each role sees a dashboard tuned to their decisions. The admin sees company-wide KPIs - total SRs, SLA compliance, revenue, NPS. The operations manager sees their queue - pending approvals, QC items, overdue jobs, team workload. The team lead sees their team - who is where, what is due, who is overloaded. The technician sees their own stats - jobs completed, average on-site time, client ratings. The finance user sees billing - outstanding quotes, unpaid invoices, expense totals.

Core KPIs tracked

- SR volume: total created, by status, by category, by site, by client, over time
- SLA compliance: percentage of SRs resolved within the target response and resolution times, breach count, average breach duration
- First-time fix rate: percentage of SRs completed without rework
- Average response time: from SR creation to technician arrival (punch-in)
- Average resolution time: from SR creation to completion
- Technician utilisation: percentage of available hours spent on active jobs

- Average on-site time: per technician, per category, per site
- Customer satisfaction: NPS score, star ratings, feedback sentiment, by technician, by site, by category
- PM compliance: scheduled vs completed, on-time vs overdue, by site, by category
- Financial: revenue by client, revenue by category, expense ratio, contract profitability
- No-show and cancellation rate

Exportable reports

- **Monthly SLA report (PDF):** Branded report for the client showing SR volumes, response times, resolution times, SLA compliance, and NPS. This is the report building managers forward to their owners.
- **Technician performance scorecard (PDF):** Per-technician monthly summary - jobs completed, first-time fix rate, average on-site time, client ratings, expense totals.
- **SR completion certificate (PDF):** Per-SR document with before/after photos, GPS data, timestamps, technician details, customer signature, and expense breakdown. Generated after QC pass.
- **Expense summary (Excel):** By project, by category, by technician, by date range, with subtotals and grand total.
- **SR data export (Excel):** Full SR list with all fields, filterable and sortable, for ad-hoc analysis.
- **Contract performance report (PDF):** Per-contract SLA compliance, calls used vs included, service cost vs contract value.

Scheduled reports: Admin can configure reports to be auto-generated and emailed on a schedule - weekly SLA summary every Monday, monthly performance scorecard on the 1st, daily expense digest. No manual intervention needed.

Module 10: Client portal

The client - the company receiving the service - gets their own login to track everything without calling the front desk.

Client authentication: Magic link login - the client enters their email, receives a login link, and is authenticated for 24 hours. No password to remember or reset. Alternatively, traditional email/password login for clients who prefer it. Each client user has a role: client admin (sees everything for their company) or client viewer (sees specific sites only).

Client dashboard: Open SRs across all their sites with status chips and last update timestamp. Completed SR history with completion dates and ratings. SLA performance against their contract - response time vs target, resolution time vs target, breach count. Quick stats - total SRs this month, average resolution time, pending items.

Raise a new SR: Simplified form - select their site, select the service category, describe the issue, attach photos. No priority selection (that is the service company's decision). No internal fields visible. On submit, the SR enters the normal Pending → Approval workflow.

View quotes and invoices: All quotes sent to them, with accept/reject buttons (same as the WhatsApp link, but inside the portal). All invoices with download links. Payment status visible.

Asset visibility: The client can see the list of assets at their sites (if the service company has enabled asset visibility for them), including warranty status and upcoming PM schedules. This builds transparency - the client sees that their equipment is being maintained proactively.

SLA performance dashboard: A visual report showing how the service company is performing against the contract - response time trends, resolution time trends, breach percentage, NPS score over time. This is the data that keeps contracts renewing.

Document access: The client can download SR completion certificates, monthly SLA reports, and contract documents - all generated by the platform and branded with the service company's identity.

Configurable forms and workflows

This is what makes one platform serve every industry without code changes.

Dynamic service forms: Each service category can have a custom checklist or data capture form that the technician fills during the job. The admin defines the form fields - text inputs, number inputs, dropdowns, checkboxes, photo captures, yes/no toggles, date pickers, and signature fields. The form is version-controlled - changing a form template does not alter the data already captured on completed SRs.

Form examples by industry

- AC service: compressor pressure, refrigerant level, filter condition (dropdown: clean/dirty/replaced), thermostat reading, drain line status
- Elevator inspection: cable condition rating, brake test (pass/fail), door alignment measurement, emergency phone test (pass/fail), safety certificate expiry date
- Generator maintenance: oil level, coolant level, battery voltage, runtime hours reading, load test result
- Fire safety: extinguisher pressure gauge reading, last refill date, signage condition, exit light test
- Cleaning service: area cleaned (checklist), completion photos, supervisor sign-off

Approval chain configuration: Different service types may require different approval flows. A routine PM job might auto-approve and go straight to assignment. A high-cost repair might require manager approval plus finance sign-off. An emergency might bypass approval entirely. The admin configures approval rules per service category and priority combination.

Custom status workflows (future): For industries with unique processes, the admin can define additional status stages or skip existing ones. A cleaning company might not need the quotation flow at all. A consulting firm might need a "report writing" stage between field visit and completion. This is a Phase 3 feature - the current 9-status workflow covers 90% of industries as-is.

Workflow automation rules: If-then rules that fire automatically. Examples: if priority is Critical AND category is Electrical, auto-assign to the electrical team lead. If a PM job is overdue by more than 7 days, escalate to the operations manager. If an SR has been in Assigned status for more than 48 hours without the technician accepting, send a reminder. These rules are configured by the admin through a UI, not hard-coded.

Inventory and spare parts

Field technicians use parts. Tracking what was used, where it came from, and what needs restocking closes the last operational gap.

Parts catalogue: A master list of all spare parts and consumables the company stocks - part name, SKU/part number, description, unit cost, preferred supplier, reorder threshold, and current stock level. Searchable and filterable.

Stock tracking: Each part has a quantity on hand. When a technician logs a part used on an SR (through the expense logging during field execution), the stock level decreases automatically. When parts are received from a supplier, the stock level increases. Stock is tracked per warehouse or per vehicle (if technicians carry stock in their vans).

Parts on SRs: During field execution, the technician selects parts from the catalogue (instead of free-typing expense descriptions). The system records: which part, how many, from which stock location, and the cost. This data flows into the invoice line items automatically - no re-entry by finance.

Reorder alerts: When a part's stock falls below its reorder threshold, the system generates an alert for the admin or the procurement team. A dashboard widget shows all parts below threshold with quantity on hand, reorder level, and preferred supplier.

Parts consumption reports: Which parts are used most, on which service categories, at which sites, by which technicians. This helps the company negotiate bulk pricing with suppliers, identify equipment that consumes unusual amounts of parts (a sign it needs replacement), and ensure technicians are not over-using consumables.

Warranty parts tracking: Some parts are covered under the equipment's warranty or service contract. The system can flag when a part used on a warranted asset should be claimed from the OEM rather than charged to the service company's own stock.

This is a Phase 2 feature. The basic expense logging (free-text with receipt photos) serves the immediate need. The full parts catalogue with stock tracking and auto-deduction is built once the core modules are stable and the first clients are using the system.

Industry adaptability matrix

The same platform serves different industries by changing configuration, not code. Here is how each module applies across ten target industries.

Columns: FM, HVAC, MEP, Equipment, Property, IT infra, Cleaning, Biomedical, Solar, Elevator

Module	FM	HVAC	MEP	Equip	Prop	IT	Clean	Bio	Solar	Elev
Reactive SRs	Core	Core	Core	Core	Core	Core	Core	Core	Core	Core
Preventive maintenance	High	High	Medium	High	Medium	Medium	High	High	High	High
Asset registry	High	Medium	Medium	Critical	Low	High	Low	Critical	High	Critical
Contract/AMC	Critical	High	High	High	Medium	Medium	Low	High	High	High

Module	FM	HVAC	MEP	Equip	Prop	IT	Clean	Bio	Solar	Elev
Configurable forms	High	High	High	High	Low	Medium	Medium	Critical	High	High
Inventory/parts	Medium	High	Medium	Critical	Low	Medium	Low	High	Medium	High
Client portal	High	Medium	Medium	Medium	High	Medium	Medium	High	Medium	Medium
WhatsApp comms	Critical	High	High	High	High	Medium	High	Medium	High	High
GPS/photo proof	Critical	High	High	High	High	Medium	Critical	Medium	High	High

Critical = the company cannot use the platform without this. **High** = they will ask for it in the first month. **Medium** = nice to have. **Low** = rarely needed.

Industry-specific configurations

- **Facility management:** Multi-site AMCs, monthly/quarterly PM schedules, subcontractor management, multi-trade SRs (one request spawns electrical + plumbing + HVAC sub-tasks), building-level asset registries, SLA reports for building owners.
- **HVAC and mechanical:** Seasonal pre-cooling/heating campaigns, refrigerant tracking forms, compressor health checklists, warranty-to-AMC conversion pipeline, equipment installation as a service type.
- **MEP contracting:** Defect liability period (DLP) tracking, snag list management, handover documentation, multi-discipline team coordination, completion certificates per trade.
- **Equipment services:** Serial-number-level tracking, usage-based PM triggers (every 1,000 runtime hours), parts consumption per asset, warranty claims against OEMs, depot repair tracking for equipment returned to the workshop.
- **Property maintenance:** Tenant request portals, common area inspection checklists, seasonal maintenance calendars, vendor/subcontractor assignments, strata/community association reporting.
- **IT infrastructure:** Server room inspection forms, CCTV/access control system checklists, network uptime tracking, asset configuration management, remote diagnostic integration.
- **Cleaning and pest control:** Visit verification (GPS + photo of cleaned area), route-based scheduling, recurring daily/weekly visit plans, client sign-off per visit, coverage area management.
- **Biomedical equipment:** Regulatory compliance checklists, calibration schedules with certificate generation, equipment uptime tracking for hospitals, strict SLA enforcement with penalty clauses, audit trail for regulatory inspectors.
- **Solar energy:** Panel performance monitoring forms, inverter inspection checklists, weather-correlated maintenance schedules, generation output tracking per site, warranty management across panel manufacturers.
- **Elevator maintenance:** Statutory inspection schedules, load test forms, safety certificate management, trapped-passenger emergency response workflow, controller firmware version tracking.

Mobile app integration

The mobile app is the technician's primary tool. It must be faster and more focused than the web portal.

Who uses it: Field technicians (Maintenance Leads) and team leads (Service Engineers). Managers and admin stay on the web portal - they need the full interface, not a mobile-optimised subset.

Core screens (Phase 1 - launch)

1. Login with biometric/PIN (after initial email + password)
2. Job pipeline - today's assigned jobs with status chips, sorted by ETA
3. Job detail - client, site, asset, description, history, contact info, map link
4. Punch in - one tap, GPS auto-captured, timestamp recorded
5. Photo capture - camera opens directly, before/after tagging
6. Service form - the configurable checklist for this service category
7. Expense logging - category picker, amount, receipt photo
8. Customer signature - full-screen signature pad
9. Punch out - locked until proof complete, same as web
10. Notifications - push notification inbox

Extended screens (Phase 2 - post-launch)

1. Asset QR scanner - scan to pull up asset details or create an SR
2. PM calendar - upcoming scheduled jobs
3. Team workload (for team leads) - see their team's jobs for the week
4. Expense history - past expenses submitted
5. Performance stats - personal metrics (jobs done, ratings, SLA)

Offline capability: The app must work in areas with poor connectivity. Job list, job details, and blank forms are cached locally. Photos, signatures, form data, and punch events are queued and sync when connectivity returns. A visible sync indicator shows pending uploads. The technician sees exactly the same job data whether they are online or offline - the only difference is that syncing pauses.

Push notifications: Firebase Cloud Messaging for Android, APNs for iOS. Events: new job assigned, schedule changed, urgent dispatch, QC result (pass/rework), daily summary available.

API dependency: The mobile app consumes the same REST API as the web frontend. No separate backend, no duplicated business logic. The API handles authentication, data retrieval, file uploads, status transitions, and push token registration.

API-first design

Every feature exists as an API endpoint. The web views and the mobile app are both consumers of the same API. Third-party systems integrate through the same endpoints.

Authentication: Sanctum token-based auth. POST /api/login returns a bearer token scoped to the user's role. Tokens have configurable expiry. Refresh tokens for mobile app sessions.

Core endpoint groups

- `/api/service-requests - CRUD, status transitions, filters by role/ownership/status/date/client/site/asset`
- `/api/assets - CRUD, search by code/serial/site, service history per asset`
- `/api/contracts - CRUD, coverage check, renewal pipeline`
- `/api/preventive-schedules - CRUD, calendar data, compliance stats`
- `/api/punch - in, out, photos, signature, expenses, hold/resume`
- `/api/quotes - create, send, client response`
- `/api/invoices - create, send, payment recording`
- `/api/dashboard - role-aware summary statistics`
- `/api/notifications - list, mark read, preferences`
- `/api/clients - CRUD, search, contacts`
- `/api/projects - CRUD by client`
- `/api/users - CRUD, role assignment, permission management`
- `/api/master-data - categories, domains, priorities, SLA matrix`
- `/api/reports - generate and download PDF/Excel reports`

Webhook events: The platform fires webhook callbacks on configurable events - sr.created, sr.status_changed, sr.completed, punch.in, punch.out, invoice.created, payment.received, contract.expiring. Enterprise clients register their webhook URL and receive real-time JSON payloads. This allows ERP systems to react to service events without polling.

API versioning: All endpoints are prefixed with /api/v1/. Breaking changes go into /api/v2/. Existing integrations are never broken by platform updates.

Rate limiting: Per-tenant rate limits to prevent abuse and ensure fair usage across SaaS tenants. Configurable by plan - Starter gets 100 requests/minute, Professional gets 500, Enterprise gets 2,000.

Documentation: Auto-generated API docs (using Scribe or Swagger) hosted at docs.aiywah.com/api. Every endpoint documented with request/response schemas, authentication requirements, and example payloads.

SaaS multi-tenancy

The platform serves multiple companies on one infrastructure, each completely isolated.

Tenant isolation (shared database, tenant_id scoping): Every data table carries a tenant_id column. A BelongsToTenant Eloquent trait adds a global scope to every query - read, update, and delete. A companion trait auto-stamps tenant_id on every insert. Cross-tenant data access is architecturally impossible - the scope is applied at the model level, not the controller level, so no developer mistake can accidentally expose another tenant's data.

Subdomain routing: Each tenant gets {slug}.aiywah.com. Middleware resolves the tenant from the subdomain at the start of every request. Unknown subdomains return a branded 404. Suspended tenants

see a "contact support" page. Enterprise tenants can use custom domains (portal.acmeservices.com) via DNS CNAME.

Subscription plans

Plan	Users	SRs/month	Key features
Starter	Up to 10	100	Basic dashboards, punch in/out, email notifications, no SE role
Professional	Up to 50	500	WhatsApp notifications, SE role, quotes and invoices, SLA tracking, daily summaries, PM scheduling
Enterprise	Unlimited	Unlimited	Own WhatsApp number, custom domain, white-label branding, REST API access, configurable forms, inventory module

Feature gating: A features JSON column on the plans table controls which modules are available. Middleware checks tenant.can('feature_name') before allowing access. Blade directives like @feature('preventive_maintenance') conditionally show UI elements. Attempting to access a gated feature shows an upgrade prompt.

Tenant provisioning (automated): When a company signs up, a queued job creates the tenant record, generates the subdomain, clones master data templates (default service categories, priorities, SLA matrix), creates the first Super Admin user with a forced password reset, and sends the welcome email. Time from signup to first login: under 60 seconds.

Platform admin panel: A management dashboard above all tenants - visible only to Aiywah FSM's own team. Tenant list with subscription status, usage metrics, MRR tracking, churn indicators. Tenant impersonation for customer support (admin logs into a tenant's workspace to debug issues). Suspension and termination controls.

Billing: Laravel Cashier with Stripe (global) or regional payment gateways (Tap for GCC, Razorpay for India). Monthly and annual billing cycles. Trial periods. Plan upgrades and downgrades with prorated billing. Automated invoice generation.

Data, security, and compliance

Service data is sensitive - client locations, equipment details, technician movements, financial records. The platform must be trustworthy at every layer.

Role-based access control (RBAC): 25+ permission keys, 7 access levels, dynamic grants. Every screen, every action, every data query is filtered by the user's permissions. Record-level security (RLS) ensures users see only records they own or are assigned to. Admin can extend or restrict any user's access without code changes.

Full audit trail: Every action is logged - who did it, when, what changed (old value → new value), from which IP address and device. The activity log is immutable (append-only, no edits or deletes by any user). Audit data is retained for a configurable period (default: 3 years). This meets the audit requirements of regulated industries like biomedical and government.

Data encryption: All data encrypted in transit (TLS 1.3). Sensitive fields (WhatsApp API credentials, payment gateway keys) encrypted at rest using Laravel's built-in encryption with per-tenant keys. File storage on S3/R2 uses server-side encryption.

Authentication security: Bcrypt password hashing. Rate limiting on login endpoints (5 attempts per minute). Forced password reset on first login for provisioned users. OTP-based password recovery via WhatsApp for field staff. Optional: two-factor authentication (TOTP) for admin and manager roles.

Tenant isolation guarantees: Global Eloquent scope (BelongsToTenant) is the architectural guarantee - it is impossible to write a query that returns another tenant's data without deliberately bypassing the scope. Database-level composite unique indexes include tenant_id to prevent ID collisions. File storage paths are tenant-prefixed. Session data is tenant-scoped.

Data residency: For Enterprise tenants with data residency requirements (government, banking, healthcare), the platform supports deployment in specific cloud regions - UAE (AWS Bahrain), India (AWS Mumbai), EU (AWS Frankfurt). The SaaS platform defaults to the nearest available region.

API security: Sanctum token auth with configurable expiry. Per-tenant rate limiting. Request validation on every endpoint. CORS configuration for approved origins only. Webhook payloads are signed so recipients can verify authenticity.

Compliance readiness: The platform does not claim specific certifications, but its architecture supports compliance with: GDPR (data export, deletion on request, consent tracking), SOC 2 (audit trails, access controls, encryption), and industry-specific requirements (medical device service records, fire safety inspection documentation, elevator safety certificates). Actual certification is a business decision, not a technical limitation.